Solution Requirements

Date	25 June 2026
Team ID	xxxxxx
Project Name	**OptiCrop – AI-Based Crop Recommendation System**.
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviours, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviours (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S. No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	User Input	Users can enter soil nutrients (N, P, K) and environmental conditions for crop prediction.	High
2	Crop Recommendation	The system predicts the most suitable crop using the trained Machine Learning model.	High

3	Data Processing	The system validates and preprocesses user inputs before making predictions.	Medium
4	Flask Backend Integration	The system receives the applicant details, validates the data, processes it through the ML model, and stores the prediction result.	High
5	Prediction Result Display	The system validates and preprocesses user inputs before making predictions	High
6	Machine Learning Model.	The application uses a trained Scikit-learn model to generate accurate crop recommendations.	Low
7	Deployment	The application is deployed on Render and accessible through a public URL.	High
8	Future Enhancements	The system should support adding new crops, datasets, and improved ML models.	Low

Step 2: Non-Functional Requirements (NFR)

S. NO	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should generate crop recommendations quickly.	Response time less than 3 seconds.
2	Scalability	The system should support multiple users simultaneously.	Supports future expansion.
3	Security & Data Privacy	User inputs should be processed securely without unauthorized access.	Secure handling of user data
4	Reliability & Availability	The application should provide accurate and consistent crop recommendations.	Available 99% of the time.
5	Usability & Accessibility	The interface should be simple, responsive, and easy to usee.	Easy to use with clear navigation.
6	Others	The system should allow easy updates to the dataset and Machine Learning model.	New crops and models can be added with minimal changes